

Mini Project Daily Work Log

Earthquake Stabilised Dialysis System for Patient Safety During Seismic Events

Duration: June 1, 2026 - June 13, 2026

Day 1 - June 1, 2026

Planned Activity: Project Title Finalization

Expected Deliverable: Approved Project Title

Work Done / Explanation:

On this day, the project title was finalized and approved after reviewing the relevance of the chosen topic to current real-world challenges. The selected title, "Earthquake Stabilised Dialysis System for Patient Safety During Seismic Events," was chosen to address the critical need for protecting dialysis patients during medical procedures when an earthquake or sudden ground vibration occurs. The title was reviewed against the project scope, feasibility, and innovation criteria before being formally approved by the project guide.

Day 2 - June 2, 2026

Planned Activity: Requirement Gathering

Expected Deliverable: Problem Statement

Work Done / Explanation:

Requirement gathering was carried out by studying existing dialysis systems and identifying the safety gaps that arise during seismic activity. It was observed that sudden tremors can displace dialysis machines, dislodge tubing connections, and interrupt blood flow, posing serious risks to patients undergoing treatment. Based on this analysis, the problem statement was framed as: 'Existing dialysis machines lack a mechanism to detect seismic disturbances and automatically stabilize or safely halt the dialysis process, putting patient safety at risk during earthquakes.'

Day 3 - June 3, 2026

Planned Activity: User & Module Identification

Expected Deliverable: Project Objectives

Work Done / Explanation:

The system's end users were identified, including dialysis patients, nursing staff, hospital technicians, and administrators. Based on these user roles, the core modules of the system were outlined: a vibration/seismic sensor monitoring module, an automatic machine stabilization and shut-off module, an alert and notification module for staff, and a monitoring dashboard module. The project objectives were defined to include real-time seismic detection, automated safety response, patient data logging, and staff alert generation.

Day 4 - June 4, 2026

Planned Activity: Use Case Diagram Preparation

Expected Deliverable: Module List

Work Done / Explanation:

A use case diagram was prepared to represent the interactions between system actors (patient, nurse, technician, administrator) and the system functions. Use cases such as 'Monitor Vibration Levels,' 'Trigger Auto-Stabilization,' 'Send Emergency Alert,' 'View Patient Dialysis Status,' and 'Generate Incident Report' were mapped. From this exercise, a finalized module list was prepared, covering the sensor module, control module, alert module, dashboard module, and reporting module.

Day 5 - June 5, 2026

Planned Activity: Database Requirement Analysis

Expected Deliverable: CRUD APIs

Work Done / Explanation:

The database requirements were analyzed to determine what data the system needs to store and manage, such as patient details, dialysis session records, sensor readings, alert logs, and staff information. Based on this analysis, the required CRUD (Create, Read, Update, Delete) operations were planned for each entity, for example, creating new patient records, reading real-time sensor data, updating dialysis session status, and deleting outdated logs. This formed the basis for the API design in later backend development stages.

Day 6 - June 6, 2026

Planned Activity: ER Diagram Design

Expected Deliverable: Table List

Work Done / Explanation:

An Entity-Relationship (ER) diagram was designed to visually represent the relationships between key entities such as Patient, DialysisSession, SensorReading, Alert, Staff, and Machine. Relationships like 'Patient undergoes DialysisSession,' 'Machine generates SensorReading,' and 'Alert is linked to Staff' were mapped with appropriate cardinalities. From this ER diagram, a finalized list of required database tables was prepared: Patient, Staff, Machine, DialysisSession, SensorReading, and AlertLog.

Day 7 - June 7, 2026

Planned Activity: Database Schema Creation

Expected Deliverable: SQL Schema

Work Done / Explanation:

Using the finalized table list and ER diagram, the actual SQL schema was created. Each table was defined with appropriate columns, data types, primary keys, and foreign key relationships. For instance, the SensorReading table includes columns for vibration intensity, timestamp, and a foreign key reference to the Machine table, while the AlertLog table links to both Staff and DialysisSession tables. The schema was created using MySQL and tested for proper relational integrity and constraints.

Day 8 - June 8, 2026

Planned Activity: UI Wireframe Design

Expected Deliverable: Page Layouts

Work Done / Explanation:

Wireframes for the major screens of the application were sketched out, including the login page, patient dashboard, real-time monitoring screen showing vibration levels, alert notification panel, and the dialysis session history page. These wireframes focused on layout structure, navigation flow, and placement of key information such as live sensor readings and emergency stop controls, without focusing on final styling. This step finalized the page layouts that would guide the detailed UI design in the following days.

Day 9 - June 9, 2026

Planned Activity: Login & Dashboard UI Design

Expected Deliverable: UI Screens

Work Done / Explanation:

Detailed UI screens for the login page and the main dashboard were designed based on the wireframes created earlier. The login screen was designed with role-based access (admin, nurse, technician), while the dashboard was designed to display real-time machine status, current vibration/seismic readings, active dialysis sessions, and quick access to alert logs. Color coding (green for normal, yellow for warning, red for emergency stabilization triggered) was incorporated to make the dashboard intuitive for medical staff during emergencies.

Day 10 - June 10, 2026

Planned Activity: Navigation & Form Design

Expected Deliverable: UI Screens

Work Done / Explanation:

Navigation menus and input forms for the application were designed, including the patient registration form, dialysis session creation form, machine configuration form, and the alert acknowledgment form. A consistent navigation bar/sidebar was designed to allow staff to move easily between the dashboard, patient records, machine monitoring, and reports sections. These screens were designed keeping in mind quick access during emergency situations, with minimal

clicks required to reach critical controls.

Day 11 - June 11, 2026

Planned Activity: Design Review

Expected Deliverable: UI Prototype

Work Done / Explanation:

A complete review of all designed UI screens was conducted, combining the login page, dashboard, navigation, and forms into a unified clickable UI prototype. Feedback was gathered on usability, color schemes, alert visibility, and overall flow between screens. Necessary corrections such as improving the visibility of emergency alerts and simplifying the patient session form were identified and noted for implementation. The finalized UI prototype was prepared as a reference for the frontend development phase.

Day 12 - June 12, 2026

Planned Activity: Frontend Environment Setup

Expected Deliverable: Design Approval

Work Done / Explanation:

The frontend development environment was set up by installing Node.js, configuring the project with a React-based build tool, and setting up the folder structure for components, pages, and assets. Required libraries for routing, state management, and UI styling were installed and configured. Alongside this setup, the finalized UI designs from the previous day's review were presented and formally approved by the project guide, marking the completion of the design phase.

Day 13 - June 13, 2026

Planned Activity: Frontend Environment Setup (Continued)

Expected Deliverable: React Project Setup

Work Done / Explanation:

Building on the previous day's environment setup, the React project structure was finalized. Base components such as the Navbar, Sidebar, Footer, and Layout wrapper were created. Routing was configured for the major pages (Login, Dashboard, Patient Records, Monitoring, Alerts, Reports) using React Router. Basic placeholder pages were created for each route, and the project was verified to run successfully on the local development server, completing the React project setup needed before actual page development begins.